

AtliQ Grands Hospitality Performance Analysis – Power BI Dashboard

1. Project Overview

This project focuses on **AtliQ Grands**, a chain of five-star hotels across India that has been experiencing a decline in market share and revenue due to increased competition and ineffective decision-making.

The goal was to develop a comprehensive Power BI dashboard to analyze key hospitality metrics and provide actionable insights for improving business performance.

2. Business Problem / Objectives

AtliQ Grands' management required a data-driven solution to understand declining performance and identify improvement areas.

The analysis aimed to answer:

- What factors are impacting revenue and occupancy?
- How efficiently are rooms being utilized?
- Which cities, properties, or room types perform best?
- How can booking performance and realization be improved?

3. Dataset Description

- **Domain:** Hospitality (Hotel Management)
- **Data Source:** Provided as part of analytics challenge
- **Data Structure:** 5 relational tables

Key Tables:

- **dim_date** – Date, week number, weekday/weekend
- **dim_hotels** – Hotel details (city, category)
- **dim_rooms** – Room types and classes
- **fact_aggregated_bookings** – Capacity & successful bookings
- **fact_bookings** – Detailed booking transactions

4. Data Cleaning & Preparation (Power Query)

- Data was relatively structured and required minimal cleaning
- Performed transformations using **Power Query**:
 - Data type corrections

- Handling inconsistencies in booking status
- Ensured consistency across date fields
- Validated revenue logic:
 - Adjusted **revenue realized** based on booking status
 - (e.g., partial refund for cancellations)

5. Data Modeling & DAX

Data Modeling:

- Built relationships across all dimension and fact tables
- Created a structured model for time-based and property-level analysis

Key KPIs Created:

- **RevPAR (Revenue per Available Room)**
- **ADR (Average Daily Rate)**
- **Occupancy %**
- **Realization %**
- **DSRN (Daily Sellable Room Nights)**
- **DBRN (Daily Booked Room Nights)**
- **DURN (Daily Utilized Room Nights)**

Additional Metrics:

- Revenue Generated vs Realized
- Booking Status Analysis (Cancelled / No Show / Checked Out)
- Ratings Analysis

6. Dashboard Overview

Developed an **interactive Power BI dashboard** for executive-level insights.

Key Analysis Areas:

1. Revenue & Performance Overview

- RevPAR, ADR, and total revenue trends
- Comparison across cities and hotel categories

2. Occupancy & Utilization

- Occupancy % trends
- Room utilization efficiency (DSRN vs DBRN vs DURN)

3. Booking Analysis

- Booking status breakdown (Cancelled, No Show, Checked Out)

- Platform-wise booking performance

4. Property & City Performance

- Performance comparison across cities
- Top and underperforming properties

5. Customer Experience

- Ratings analysis
- Impact of service quality on bookings and revenue

7. Dashboard & Data Screenshots

Figure 1: CSV files (Data sources)

The figure displays three Excel spreadsheets representing data sources for a hotel booking dashboard.

fact_aggregated_bookings

property_id	check_in_date	room_category	successful_bookings	capacity
16558	01-May-22	RT1	25	30
19562	01-May-22	RT1	28	30
19563	01-May-22	RT1	23	30
17558	01-May-22	RT1	13	19
16558	01-May-22	RT1	18	19
17560	01-May-22	RT1	28	40
16558	01-May-22	RT1	25	40
19560	01-May-22	RT1	23	26
17561	01-May-22	RT1	22	26
16560	01-May-22	RT1	24	34
16561	01-May-22	RT1	16	18
19562	01-May-22	RT1	20	31
16563	01-May-22	RT1	36	41
17559	01-May-22	RT1	26	32
17562	01-May-22	RT1	12	20
17563	01-May-22	RT1	21	25
18558	01-May-22	RT1	11	15
18559	01-May-22	RT1	29	42
18561	01-May-22	RT1	31	33

dim_hotels

property_id	property_name	category	city
16558	Atiq Grands	Luxury	Delhi
16559	Atiq Exotica	Luxury	Mumbai
16560	Atiq City	Business	Delhi
16561	Atiq Blu	Luxury	Delhi
16562	Atiq Bay	Luxury	Delhi
16563	Atiq Palace	Business	Delhi
17558	Atiq Grands	Luxury	Mumbai
17559	Atiq Exotica	Luxury	Mumbai
17560	Atiq City	Business	Mumbai
17561	Atiq Blu	Luxury	Mumbai
17562	Atiq Bay	Luxury	Mumbai
17563	Atiq Palace	Business	Mumbai
18558	Atiq Grands	Luxury	Hyderabad
18559	Atiq Exotica	Luxury	Hyderabad
18560	Atiq City	Business	Hyderabad
18561	Atiq Blu	Luxury	Hyderabad
18562	Atiq Bay	Luxury	Hyderabad
18563	Atiq Palace	Business	Hyderabad
19558	Atiq Grands	Luxury	Bangalore

fact_bookings

booking_id	property_id	booking_date	check_in_date	checkout_date	no_guests	room_category	booking_platform	ratings_given	booking_status	revenue_generated	revenue_realized
May012216558RT11	16558	27/04/2022	01/05/2022	02/05/2022	3	RT1	direct online	1	Checked Out	10010	10010
May012216558RT12	16558	30/04/2022	01/05/2022	02/05/2022	2	RT1	others	1	Cancelled	9100	3640
May012216558RT13	16558	28/04/2022	01/05/2022	04/05/2022	2	RT1	logtrip	5	Checked Out	9100	9100
May012216558RT14	16558	28/04/2022	01/05/2022	02/05/2022	2	RT1	others	1	Cancelled	9100	3640
May012216558RT15	16558	27/04/2022	01/05/2022	02/05/2022	4	RT1	direct online	5	Checked Out	10920	10920
May012216558RT16	16558	01/05/2022	01/05/2022	03/05/2022	2	RT1	others	4	Checked Out	9100	9100
May012216558RT17	16558	28/04/2022	01/05/2022	06/05/2022	2	RT1	others	1	Cancelled	9100	3640
May012216558RT18	16558	30/04/2022	01/05/2022	05/05/2022	2	RT1	logtrip	1	No Show	9100	9100
May012216558RT19	16558	30/04/2022	01/05/2022	02/05/2022	2	RT1	tripster	1	Checked Out	9100	9100
May012216558RT110	16558	28/04/2022	01/05/2022	02/05/2022	1	RT1	others	4	Checked Out	9100	9100
May012216558RT111	16558	29/04/2022	01/05/2022	07/05/2022	1	RT1	makeyourtrip	5	Checked Out	9100	9100
May012216558RT112	16558	26/04/2022	01/05/2022	02/05/2022	2	RT1	logtrip	5	Checked Out	9100	9100
May012216558RT113	16558	26/04/2022	01/05/2022	07/05/2022	2	RT1	makeyourtrip	5	Checked Out	9100	9100
May012216558RT114	16558	30/04/2022	01/05/2022	05/05/2022	1	RT1	logtrip	1	Cancelled	9100	3640
May012216558RT115	16558	29/04/2022	01/05/2022	02/05/2022	2	RT1	others	1	Checked Out	9100	9100
May012216558RT116	16558	27/04/2022	01/05/2022	03/05/2022	4	RT1	journey	4	Checked Out	10920	10920
May012216558RT117	16558	29/04/2022	01/05/2022	02/05/2022	3	RT1	direct offline	1	Checked Out	10010	10010
May012216558RT118	16558	27/04/2022	01/05/2022	02/05/2022	2	RT1	others	1	Checked Out	9100	9100
May012216558RT121	16558	07/04/2022	01/05/2022	06/05/2022	2	RT2	others	1	Checked Out	12900	12900

Occupancy & Room Utilization

- Gaps observed between **available rooms (DSRN)** and **utilized rooms (DURN)**
- Indicates underutilization of capacity

Recommendation:

Optimize pricing strategies and promotional offers to increase room occupancy and utilization.

Booking Behavior

- A significant portion of bookings fall under **cancellations and no-shows**, impacting realized revenue

Recommendation:

Implement stricter cancellation policies and incentives for confirmed stays to reduce revenue leakage.

Property & City Performance

- Performance varies across cities and properties, with some locations underperforming

Recommendation:

Identify low-performing properties and improve marketing, pricing, or service quality in those locations.

Customer Experience

- Customer ratings directly influence booking trends and repeat business

Recommendation:

Enhance service quality and customer experience to improve ratings and long-term revenue.

9. Tools & Technologies Used

- Power BI – Dashboard development & DAX
- Power Query – Data transformation
- Excel / CSV – Data source

10. Key Skills Demonstrated

- Data modeling with multiple fact & dimension tables
- KPI development using DAX
- Business performance analysis (hospitality domain)
- Dashboard design & storytelling
- Translating data into actionable insights